

Coding Section

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Pattern Based Coding Questions

<https://www.edureka.co/blog/30-pattern-programs-in-java/>

a. Pyramid Pattern

a. Diamond Pattern

Hint: Pyramid's upper part code will remain the same only lower part logic code needs to add

a. Left Triangle Pattern

Hint: If we remove space code block from Pyramid pattern code then left triangle pattern will print

a. Right Triangle Pattern

Hint: If we remove little space from "*" from pyramid source code then it will print right triangle pattern

Reg Ex Based coding Questions

<https://www.hackerrank.com/domains/regex>

<https://runestone.academy/ns/books/published/py4e-int/regex/hp-practice-adv.html>

<https://www.interviewbit.com/coding-interview-questions/>

Coding Questions should be know before every interview

Important Import packages in Java before starting

```
import java.util.List;
import java.util.ArrayList;
import java.util.Iterator;
import java.util.Map;
import java.util.HashMap;
import java.util.LinkedHashMap;
import java.util.stream.Collectors;
import java.util.Collections;
import java.util.Optional;
```

To find min and max in Collections object

<https://www.geeksforgeeks.org/finding-minimum-and-maximum-element-of-a-collection-in-java/>

1. Iterator --> Remove element using Iterator | Done

<https://www.geeksforgeeks.org/iterators-in-java/>

1. **Stream API --> sort element based on age in map having name and age as key value pair | Done**

Why to use entrySet() ?

<https://www.naukri.com/code360/library/hashmap-entryset-method-in-java>

1. **Duplicate letter in a string | Done**

Mississippi : return letter and it's count

<https://www.javatpoint.com/java-program-to-count-occurrences-of-each-character>

<https://stackoverflow.com/questions/10462819/get-keys-from-hashmap-in-java>

<https://www.geeksforgeeks.org/hashmap-values-method-in-java/>

1. **To check palindrome --> can't reverse and store | Done**

<https://www.geeksforgeeks.org/java-program-to-check-whether-a-string-is-a-palindrome/>

Easiest and simple method using StringBuilder method

<https://medium.com/@brandon93.w/stringbuilder-and-stringbuffer-in-java-examples-and-interview-exercises-fd735f15c415#:~:text=This%20is%20an%20unsynchronized%2C%20mutable,create%20multiple%20intermediate%20string%20objects.>

1. **Code to give second highest element of the array without sorting | Done**

<https://www.geeksforgeeks.org/find-second-largest-element-array/>

1. **Reverse string (use stringbuilder or stringbuffer) --> Completed**

<https://www.geeksforgeeks.org/stringbuffer-class-in-java/>

1. **Print unique elements from the list --> Completed | Done**

- Using Stream API
- Get max and min using Stream API
- Sort it out using Stream API

1. **Second largest element from the list using java 8 stream API | Done**

1. **Using hashmap print the number of times element is repeating as value and the element as key(e.g. List of (12,12,12,14) | Done**

Key -12, value 3

Key -14, value 1

1. **Swap two numbers without using temporary variable | Done**

<https://www.w3schools.in/java/examples/swapping-two-numbers-without-using-a-temporary->

[variable](#)

1. **Distinct character count in string (Use hashmap) | Done**

1. **Prove string is immutable --> Completed | Done**

Example 1: <https://stackoverflow.com/questions/27902471/program-to-prove-string-is-immutable>

a. **Show Diamond problem with multiple inheritance | Done**

<https://www.prepbytes.com/blog/java/diamond-problem-in-java/>

1. Reverse string without in built method and sort
Hint: Iterate over string and traverse from last

1. **Employee class is there in which 3 attributes are there (id, name, salary) and return second highest salary | Done**

<https://www.jrebel.com/blog/java-streams-cheat-sheet>

<https://www.linkedin.com/pulse/how-safely-use-map-operation-java-8-streams-nikhil-gargatte/>

<https://www.digitalocean.com/community/tutorials/java-stream-collect-method-examples>

1. **Sort hashmap by value --> Completed | Done**

1. **input string --> convert into map --> 1 as key :one as value**

String input = "1:one,2:two,3:three";

2. **two integer array --> merge array --> without duplicate**

```
// array1 : 1,2,3,4  
// array2 : 4,3,7,2,1
```

Reference:

<https://stackoverflow.com/questions/61297074/using-java-streams-to-merge-a-pair-of-int-arrays>

<https://howtodoinjava.com/java8/stream-concat-example/#:~:text=Java%20%20Stream-,The%20Java%20%20Stream.,also%20be%20a%20parallel%20stream.>

3. **int[] arr = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9};**

/* Expected Output

[0, 1, 2]

[3, 4, 5]

[6, 7, 8]

[9]

*/

1. **Input:**

```
name = [ "Akshay", "Neha", "Snehar", "Vipin" ]  
roll_no = [ 4, 1, 3, 2 ]  
marks = [ 40, 50, 60, 70 ]
```

Output:

The zipped result is : {'Akshay', 4, 40}, ('Snehar', 3, 60), ('Vipin', 2, 70), ('Neha', 1, 50)}

1. **To count no of digit in Integer | Done**

- list of fruits filter out starts with 'M'
- positive no at the end and negative number at the start
- Program to parenthesis balancing**

<https://www.geeksforgeeks.org/check-for-balanced-parentheses-in-an-expression/>

- find all the triplet from the given list of no that sum up to the target**
- Common longest Substring in List of type string**

7. **Recursion problem**

<https://www.theserverside.com/blog/Coffee-Talk-Java-News-Stories-and-Opinions/examples-Java-recursion-recursive-methods>

- how you can find the employee with the highest salary using Java**
- Rotate the array by k unit**

For example, if nums = [1,2,3,4,5,6,7] and k = 3, the output should be [5,6,7,1,2,3,4].

- Count the number of palindromic substrings in a string**
- Java 8 Questions --> Important for EY**

Reference: <https://javaconceptoftheday.com/java-8-interview-sample-coding-questions/>

Given a list of strings, find out those strings which start with a number?

How do you find common elements between two arrays?

<https://www.geeksforgeeks.org/find-common-elements-in-two-arrays-in-java/>

Java 8 | How do you merge two unsorted arrays into single sorted array without duplicates?

Find the age of a person in years if the birthday has given?

<https://www.geeksforgeeks.org/how-to-calculate-age-from-a-birthdate-using-java-date-time/>

Java 8 | Given an integer array, find sum and average of all elements?

<https://www.w3resource.com/java-exercises/stream/java-stream-exercise-1.php>

Given a list of strings, sort them according to increasing order of their length?

Java 8 | Find sum of all digits of a number in Java 8?

Java 8 | Program to check if two strings are anagrams or not?

Java 8 | Find the longest string in a list of strings

Java 8 | sum of square of odd number using Stream API

`int[] arr = {1,2,3,4...10};`

Java 8 | IMP | Find the Most Frequent Character in a String (excluding spaces)

Java 8 | IMP | Group Employees by Department and List Names Alphabetically

Java 8 | IMP | FlatMap a List of Sentences into Unique Sorted Words

Java 8 | IMP | Count Word Frequency from List of Strings

Java 8 | IMP | Nested Grouping: Group by Department, Then by Age

Data structure & Algorithm Questions

<https://workat.tech/problem-solving/practice/topics>

<https://sathishsuresh.medium.com/cracking-data-structures-coding-interview-abca041dbd59>

a. LinkedList Operations

- To initialise the LinkedList
- To print the element of LinkedList

- Reverse logic
- To print the element of reverse LinkedList
- To find loop in LinkedList

<https://www.interviewbit.com/blog/detect-loop-in-linked-list/>

b. Tree Operations

- Insert elements in the Tree
- Traversal operations

Reference:

<https://www.geeksforgeeks.org/tree-traversals-inorder-preorder-and-postorder/>

c. Graph Operation

- Adjacency List (Using HashMap)
- Adjacency Matrix (2D Array)
- Weighted Graph using Adjacency List

d. Stack and Queue

- Clone Graph

e. Heap

- Sliding Window Maximum
- Kth Largest Element

f. HashMap and Set

- Top K frequent Elements
- Find duplicates in Array

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